Exploring the Snort Intrusion Detection System

Modern enterprises need IT for smooth and efficient functioning. The almost universal use of the Internet has spawned many bad actors and malicious agents. There are hackers out there waiting to break into computer systems for espionage, theft of industrial secrets and for information. Intrusion detection systems help address this problem.

An intrusion detection system (IDS) is a network security technology built to detect vulnerabilities and has the ability to block threats. According to Brian Rexroad, vice president of security platforms for AT&T, "The overall purpose of an IDS is to inform IT personnel that a network intrusion may be taking place. Alerts will generally include information about the source address of the intrusion, the target's/victim's address, and the type of attack that is suspected."

What is involved in network intrusion?

To detect and to prevent hackers and attackers from penetrating a system, we need to have a basic understanding of the attacks that can be possibly performed on it.

Scanning attack: This involves sending packets/information to a network in an attempt to gather data about the network, e.g., blind SQL injection.

Buffer overflow attack: This is an attempt to penetrate sections of the memory in the devices connected to the network, replacing the normal data with malicious code. This can cause the entire system to crash and create chaos, which helps in hiding an attack on another point in the system.

Protocol-specific attack: This targets specific protocols such as ICMP, TCP and ARP.

- *ICMP* stands for Internet Control Message Protocol; *ping floods* attack can be performed, which overwhelm the device with ICMP echo-request packets. There are also *smurf* attacks and port scanning.
- *TCP* stands for Transmission Control Protocol, which is vulnerable to *TCP syn* attacks in which a port stays open as the ACK message is never received; the open port can be used to send malicious packets.
- *ARP* stands for Address Resolution Protocol, in which the attack takes place by *ARP Poisoning* where false ARP messages are sent to link the attacker's MAC address with the IP address of a legit device.

Malware: This includes worms, trojans, viruses and bots. Basically, as the word itself says, this software is designed to damage or disrupt the system.

Traffic flooding: This is also known as DDoS attack.

Types of intrusion detection systems

Network-based intrusion detection systems (NIDS) are deployed at strategic points throughout the network, basically to keep a watch over places where the traffic is most likely to be vulnerable. They are relatively easy to secure and thus, an intruder may not realise that an attack is being detected. NIDS analyses a large volume of network traffic, which means it has low specificity. So, there are chances of it missing an attack or not detecting something in encrypted traffic.

A host intrusion detection system (HIDS) is established on all devices in the network. It works as a second line of defence against malicious data if NIDS fails to detect something. It looks at the entire